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$$\int \frac{1}{x^2 \sqrt{x^2 - 1}} dx \stackrel{x = \frac{1}{\cos^2 t}}{=} \int \frac{1}{\frac{1}{\cos^2 t} \sqrt{\frac{1}{\cos^2 t} - 1}} \frac{\sin t}{\cos^2 t} dt = \int \frac{\sin t}{\tan t} dt = \int \cos t dt$$

$$= \sin t + c \stackrel{t = \arccos\left(\frac{1}{x}\right)}{=} \frac{\sqrt{x^2 - 1}}{x} + c$$

References